

5. Factor Price Equalization and Real Incomes

Solution and Explanation:

(1) Stolper–Samuelson Theorem Recap

Definition: The **Stolper–Samuelson theorem** states that an increase in the relative price of a good raises the real return to the factor used intensively in that good and lowers the real return to the other factor.

Formally, if Computers (C) are capital-intensive:

$$\frac{P_C}{P_T} \uparrow \Rightarrow r \uparrow, w \downarrow$$

Conversely, if Textiles are labor-intensive:

$$\frac{P_T}{P_C} \uparrow \Rightarrow w \uparrow, r \downarrow$$

(2) Quantitative Derivation

We now derive how the percentage change in good prices affects factor prices.

Step 1: Unit cost functions

For each good produced with constant returns to scale and competitive markets, the price equals the unit cost:

$$P_i = a_{Ki}r + a_{Li}w, \quad i = C, T$$

where a_{Ki} and a_{Li} are input requirements per unit of output.

Taking percentage changes (denoted by hats):

$$\hat{P}_i = \theta_{Ki}\hat{r} + \theta_{Li}\hat{w}$$

where θ_{Ki} and θ_{Li} are cost shares of capital and labor in good i , satisfying $\theta_{Ki} + \theta_{Li} = 1$.

Step 2: Set up system of equations

For Computers ($i = C$) and Textiles ($i = T$):

$$\begin{cases} \hat{P}_C = \theta_{KC}\hat{r} + \theta_{LC}\hat{w} \\ \hat{P}_T = \theta_{KT}\hat{r} + \theta_{LT}\hat{w} \end{cases}$$

Using the given production exponents (which correspond to cost shares):

$$\theta_{KC} = 0.6, \quad \theta_{LC} = 0.4; \quad \theta_{KT} = 0.3, \quad \theta_{LT} = 0.7$$

Step 3: Substitute known price changes

In Canada:

$$\hat{P}_C = +0.20, \quad \hat{P}_T = -0.10$$

Substitute:

$$\begin{cases} 0.20 = 0.6\hat{r} + 0.4\hat{w} \\ -0.10 = 0.3\hat{r} + 0.7\hat{w} \end{cases}$$

Step 4: Solve the system for $\hat{r}$ and $\hat{w}$

Multiply the second equation by 2 to align the coefficients of $\hat{r}$:

$$\begin{cases} 0.20 = 0.6\hat{r} + 0.4\hat{w} \\ -0.20 = 0.6\hat{r} + 1.4\hat{w} \end{cases}$$

Subtract the second equation from the first:

$$0.40 = -1.0\hat{w} \quad \Rightarrow \quad \hat{w} = -0.40$$

Substitute into the first equation:

$$0.20 = 0.6\hat{r} + 0.4(-0.40) \Rightarrow 0.20 = 0.6\hat{r} - 0.16 \Rightarrow 0.36 = 0.6\hat{r} \Rightarrow \hat{r} = 0.60$$

Interpretation:

$$\boxed{\hat{r} = +60\%, \quad \hat{w} = -40\%}$$

Conclusion: A 20% increase in the price of Computers (and a 10% fall in Textiles) leads to: - A 60% increase in the real return to capital (r) - A 40% fall in the real wage (w)

This demonstrates that factor price changes are *magnified* relative to goods price changes — a key result of the Stolper–Samuelson theorem.

Economic meaning: - Capital owners in Canada gain substantially. - Workers lose both relatively and absolutely (their real wage falls even though overall welfare may rise).